

You are evaluating an ontology. This is a non-interactive task. Use the GPT 5.5 High Model. Return exactly one valid downloadable JSON file and nothing else.

Do not:

- ask questions;
- request clarification;
- summarize the ontology;
- explain your reasoning outside the JSON.

Task: Identify ontology modeling flaws, quality issues, pitfalls, consistency problems, and maintainability risks that are directly supported by the provided evidence.

Your goal is high recall with low duplication:

- Report every distinct ontology flaw that is sufficiently supported by the evidence.
- Do not omit valid findings simply because many similar issues exist.
- Merge repeated occurrences of the same modeling flaw into a single finding.

Evidence: Three types of evidence sources are provided (16 files).

1. Primary evidence: The original FASTO_RDF.owl in RDF/XML syntax. This is the authoritative source.
2. Secondary evidence: A rule-based verbalizer generated from a temporary sanitized copy of FASTO. This exists only to improve readability.
3. Tertiary evidences: .CSV, TXT, TSV, and Turtle files containing results from standard ontology-analysis tools (OOPS! and ROBOT), and from manual investigation (RDFLib). Use these files as an existing baseline of flaws. Do not reproduce or paraphrase their findings.

Evidence rules:

- RDF/XML is always the primary source.
- Enhance the results of the tertiary sources (results of standard tools and manual inspection), if those are not sufficient.
- The verbalizer is supplementary evidence only.
- Do not compare the RDF/XML and verbalizer.
- Do not use the verbalizer to judge datatype correctness.
- Use tertiary evidence to guide or expand the analysis.
- Do not simply repeat, paraphrase, or reformat findings already present in the supplementary files.
- Analyze all sources independently to identify additional issues that are not already captured in the supplementary files.
- Report only findings supported by the supplied evidence.
- Do not invent ontology entities, intended meanings, missing axioms, or external facts.

- If evidence is incomplete or ambiguous, report the finding only with appropriately low confidence.

Reporting rules:

A finding represents one unique ontology modeling problem, not one affected entity.

If the same modeling flaw affects multiple entities:

- create one finding;
- aggregate all affected entities;
- aggregate the supporting evidence.

Do not create separate findings that differ only by:

- entity;
- IRI;
- repeated evidence;
- wording;
- equivalent explanation.

Normal subclass specialization alone is not evidence of redundancy.

Do not infer flaws solely because subclass relationships exist.

Do not repair the ontology.

Suggested fixes must be textual recommendations only.

Output requirements

Return exactly one JSON object.

Do not add fields outside the schema.

Every finding must contain:

- id
- entity
- entity_type
- flaw_category
- error_type
- severity
- evidence
- explanation
- suggested_fix
- confidence
- additional_comment

Field rules:

entity: String or array identifying every affected ontology entity.

entity_type: Use the most appropriate ontology type (e.g., Class, ObjectProperty, DatatypeProperty, AnnotationProperty, Individual, Ontology, Multiple, Unknown).

flaw_category: Short descriptive category of the modeling flaw.

error_type One of:

- error
- warning
- info

severity One of:

- critical
- important
- minor

confidence: Numeric value between 0.0 and 1.0.

evidence: Must reference or quote the supplied input that supports the finding.

suggested_fix: Provide a textual recommendation only. Do not generate OWL, RDF/XML, Manchester syntax, or repaired axioms.

Global comments: Use global_comments only for ontology-wide observations that:

- are directly supported by the evidence;
- cannot reasonably be attached to a single finding.

Internal quality checks

Before producing the JSON, internally verify that:

- every finding is supported by evidence;
- no duplicate findings remain;
- repeated instances of the same flaw have been merged;
- every required field is present;
- the JSON is syntactically valid.

If no well-supported ontology flaw is found, return:

"findings": []

Required JSON schema

```
{
  "task": "general_ontology_flaw_detection",
  "model": "",
  "input_condition": "llm_enhanced",
  "findings": [
    {
      "id": "F1",
      "entity": "",
      "entity_type": "",
      "flaw_category": "",
      "error_type": "error",
      "severity": "important",
```

```
"evidence": "",
"explanation": "",
"suggested_fix": "",
"confidence": 0.0,
"additional_comment": ""
}
],
"global_comments": [
{
"comment": "",
"evidence": "",
"confidence": 0.0
}
]
}
```

Now analyze the supplied ontology evidence and return the downloadable JSON file only.